

ITAI 13: Introduction to Artificial Intelligence

Lab 03: Contribution Journal

Group Name: 14274 2_ITAI1371

Contributors:

Marisabel Morales – Lab Notebook & example

Joshua Balderas – Reflective Journal

Katherine Rodriguez – Contribution Journal

Date: September 6, 2026 Contribution Journal

Submitted by: Katherine Rodriguez

Introduction:

1. This journal outlines each team member's individual contributions to Lab 03. Each section documents one team member's assigned contribution in their own words.

2. Marisabel Morales – Lab Notebook Contributions

"My responsibility was documenting the lab notebook and the example."

Tasks Completed:

Created lab notebook that explained ML workflow and practice with an example(collection, cleaning, model selection, training, evaluation, deployment)

Documented data analysis (looking at dataset, making note of missing values, noting features' distributions, etc.)

Documented the code with well-written explanations for each model implemented (random forest, logistic regression, SVM)

Documented evaluation metrics and well-written explanations of each evaluation metric and when you would want to use each.

Wrote the computer vision portion which included explaining digital image representation, the computer vision pipeline, and the core tasks.

Commented all code

Reflection: I learned that writing down all of the steps that we took while working through the lab allowed me to have a better understanding of the concepts.

3. Joshua Balderas – Reflective Journal Contributions

“My responsibility was writing deeper than simply stating what I did in the lab. This included what you learned, what you struggled with, and how your thinking may have changed.”

Tasks Completed:

Reflected on ML workflow and how each step flows into the next. Importance of understanding how it works.

Reflected on what allowed you to understand the differences between supervised vs unsupervised learning.

Realized you did not understand overfitting and the picture helped.

Reflecting on using the right evaluation metric and how not every metric is good for every situation. We used the example of disease screening.

Reflection on how the article changed your thinking about AI

Wrote down questions you had while working through the lab

Sample Reflection:

“Once I took the time to think about what distinguishes supervised from unsupervised data, I realized it really depends on if you have the “answer key” when training your model. Having the “answer key” allows you to know when you’re right or wrong as you work. That was the defining moment when I understood what made these two types of learning different. If you have that “answer key”, you are learning in a supervised way. If you don’t have it, you are learning in an unsupervised way.”

Reflection: For me the most valuable part of this lab was writing the reflective journal. It really allowed me to think about what I learned and where I still had questions. Now I feel like I have a strong foundation when it comes to ML.

4. Katherine Rodriguez – Contribution Journal Contributions

“My responsibility was to manage the project/code implementation and document each team members contribution.”

Tasks Completed:

Project Management:

Created group's file naming structure according to convention.

Created group timeline to complete work by.

Assigned roles with help from Marisabel Morales and Joshua Balderas.

Created group channel to communicate and share work.

Double checked all files were named according to convention before submission.

Data Preparation:

Implemented code to load and explore data.

Cleaned data by accounting for missing values.

Created train/test split (80/20)

Scaled and normalized features.

Model Development:

Implemented model training for Random Forest, Logistic Regression, SVM.

Performed model hyperparameter tuning with GridSearchCV.

Created evaluation metrics and confusion matrices.

Documentation:

Created this entire contribution journal. Documents each team member's contributions with specific code examples.

Reflection: Having to manage the project and implement the code allowed me to get hands on experience with all parts of the ML workflow. I really enjoyed hyperparameter tuning the most. I feel like that gave me a better understanding of how you can use systematic testing to help you improve your model. What I

learned was the most valuable was how to balance both implementing code and managing a project.

5. Summary of Contributions

Team Member	Primary Role	Key Contributions
Marisabel Morales	Lab Notebook	Wrote clear descriptions of technical implementation. Explained code and wrote about computer vision concepts. Explained each evaluation metric and when you would want to use each.
Joshua Balderas	Reflective Journal	Reflections on what you learned, what you struggled with, and how your thinking changed after working through the lab.
Katherine Rodriguez	Contribution Journal	Managed team/project and implemented data prep and model development. Documented each team member's contributions.

6. Quality Assurance Checklist

Did you complete all required tasks?

Code is commented if there are complex sections

Did your journal reflect deep and genuine reflection on the learning process?

Were all files named exactly how the file naming convention specified?

Did you complete your work independently?. (I assigned roles but we collaborated to make sure everyone knew what they were doing)

7. Academic Integrity Statement:

We, the undersigned, certify that this work has been accomplished honestly and in compliance with the Academic Integrity Policy of this institution. All work submitted is the work of the author(s) unless quoted or referenced correctly. Course documents, published articles, and internet references were used as resources unless prohibited.